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TECHNICAL ARTICLE

Efficient Leaching and Selective Cobalt Extraction from Waste Lithium-Ion Battery Cathodes

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The effective recovery of valuable metals from waste lithium-ion batteries (LIBs) can promote synergistic growth in both economic and environmental domains. This study underscores the potential benefits of an environmentally sustainable approach that combines leaching and electrodeposition for the recovery of valuable metals from waste lithium-ion batteries. Specifically, it proposes a "citric acid + sodium thiosulfate" system for the effective extraction of valuable metals from waste LIBs along with the separation of Co and Li from the leaching solution via electrodeposition. High-purity $LiCoO_2$ powder was extracted following pretreatment. Optimal leaching conditions were established through one-way experiments, achieving leaching rates of 98% for both Co and Li. Kinetic studies indicate that leaching primarily involves a combination of surface chemical reactions and diffusive mass transfer. Density functional theory (DFT) investigations revealed that elemental cobalt in solution is predominantly stabilized in the form of cobalt citrate dimer. The separation of metals in the leach solution was achieved by removing cobalt metal via electrochemical analysis. The results demonstrate that the proposed method for the efficient extraction of valuable metals from waste LIBs, along with subsequent metal separation, offers a viable solution for recycling waste LIBs.

INTRODUCTION

In light of the current wave of technological innovation and the peak of lithium-ion battery decommissioning, the efficient recycling of precious metal components from waste lithium-ion batteries has become a critical issue for the new energy industry's utilization of solid waste resources.[1,2] The European Union aims to mitigate the negative environmental impacts of waste batteries effectively by promulgating and implementing the Batteries and Waste Batteries Regulation, which comprehensively covers the entire life cycle management of batteries from production to reuse to eventual recycling, as well as to protect environmental quality and human health and safety.[3–5] China has clearly articulated its ambitious strategic goal of achieving carbon peak and carbon neutrality, aiming to promote the green transformation and sustainable development of its new energy industry chain, supported by a series of forward-looking strategic policy documents such as the "14th Five-Year Plan for Industrial Green Development." [6–8] However, discarded lithium-ion batteries pose a significant challenge to our development. These batteries contain precious metals, which, if not properly recycled, result in substantial resource waste.[9–11] More importantly, if the toxic components contained in these substances leak out, it will cause irreparable environmental pollution, disrupt the ecological balance, and endanger human health.[12–14] Consequently, the proper recycling of waste lithium-ion batteries has become a primary priority.

The lithium-ion battery (LIB) system comprises several key types, including lithium cobalt oxide (LCO), lithium manganese oxide (LMO), lithium iron phosphate (LFP), and nickel-cobalt-manganese

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(NCM) batteries.[15–18] Among these, LCO batteries play a significant role, as they are widely utilized in various applications due to their excellent energy density, long cycle life, outstanding discharge performance, relatively reliable safety, high technological maturity, and steadily increasing market demand.[19,20] However, the rising popularity and widespread use of LCO batteries have led to a rapid increase in waste production, which poses serious environmental concerns and places a strain on natural ecosystems.[21,22] Consequently, the appropriate disposal of waste LCO batteries has become critical. Recycling and reusing waste LCO batteries represent a practical strategy with far-reaching implications for both environmental protection and resource recovery. Given that their cathode materials contain high-value cobalt and lithium, recycling these materials can significantly enhance the recovery of secondary resources, resulting in both economic and environmental benefits.[23–25]

Physical recycling and chemical recycling are the predominant recycling processes employed today. Physical recycling primarily focuses on recovering valuable metals from batteries through mechanical crushing and sorting. This process typically involves crushing, screening, magnetic separation, and air separation of the battery to isolate its various components, such as the casing, diaphragm, and positive and negative electrode materials. However, factors such as cell shape and component complexity can significantly influence the efficiency of physical recovery.[26–28] On the other hand, chemical recovery mainly utilizes pyrometallurgical and hydrometallurgical processes to recover materials. The pyrometallurgical method employs high-temperature molten states and carbon reductants to produce valuable metal alloys. However, this reaction necessitates high temperatures, typically around 1400 °C, leading to substantial energy consumption. Additionally, the reaction generates harmful gases and fumes that, if not properly managed, can adversely affect the environment.[29–31] To separate these components, further processing is required, including wet procedures. In contrast, hydrometallurgy offers advantages such as lower energy consumption, higher metal recovery efficiency, and improved product purity, while being less likely to produce hazardous compounds. Consequently, it is frequently employed in the recycling of lithium-ion batteries.[32–34]

In hydrometallurgy, the extraction of essential metals from cathode materials of waste lithium-ion batteries (LIBs) primarily relies on acid, alkali, and microbial leaching methods.[35–40] Inorganic acids, being abundant and cost-effective, have emerged as the most widely employed leaching agents due to their ability to effectively and rapidly dissolve various metal components, resulting in high recovery rates. Commonly utilized inorganic acids include hydrochloric acid (HCl),[41] sulfuric acid (H_2SO_4),[42] and nitric acid (HNO_3).[43] Among these, H_2SO_4 is considered a more stable and economically viable leaching agent compared to HCl and HNO_3, facilitating its industrial application. The industry typically employs a combination of H_2SO_4 and hydrogen peroxide (H_2O_2) to extract metals from waste lithium-ion battery cathode materials. This method efficiently dissolves all metals and achieves a high recovery rate.[44–46] However, despite the widespread industrial use of inorganic acids due to their superior leaching efficiency, the generation of harmful gases, such as chlorine gas (Cl_2) and sulfur trioxide (SO_3), during their application poses significant ecological risks.[47,48]

The use of organic acids for the recovery of metals from waste lithium-ion batteries is recognized as an environmentally friendly technique characterized by high efficiency, low corrosivity, and biodegradability.[49,50]

These organic acids typically include acetic acid (AA), lactic acid (LA), and gluconic acid (GA).[51–53] In the field of lithium-ion battery waste disposal, they are highly regarded for their ability to ensure safety and minimize environmental impact to the greatest extent. The reducing agent plays a crucial role in the organic acid-based metal leaching process.[54–56] However, while hydrogen peroxide (H_2O_2) is often used as a reducing agent and is effective, it poses safety risks due to its significant corrosive properties, which can lead to explosions and the release of toxic gases if not handled or stored properly.[57,58] Furthermore, organic acids tend to be more expensive and form more stable complexes than inorganic acids, necessitating more complex and delicate separation processes during recovery to ensure efficient metal recovery and purity.[59] This complexity introduces additional challenges in the subsequent metal separation stages.

In this study, a combined method of leaching (citric acid + sodium thiosulfate) and electrodeposition was employed to recover cobalt metal from waste lithium cobalt oxide (LCO). This method facilitates the efficient and direct recovery of cobalt by forming stable metal complexes during the leaching stage, which allows for the direct extraction of metals in the subsequent step. This approach eliminates the need for a complex intermediate separation process, thereby simplifying the overall procedure. Initially, the waste LCO was pretreated, and a systematic single-variable control experimental design was implemented to investigate the specific impacts of various influencing factors on metal leaching efficiency. Additionally, the system's leaching kinetics were examined, revealing the key factors limiting efficient leaching. This study employs mechanism analysis to elucidate the intrinsic mechanisms by which the citric acid and sodium thiosulfate system enhances the efficient leaching of

cobalt and lithium from the cathode materials. Finally, the leaching solution underwent electrochemical evaluation, and the electrodeposition technique was utilized to extract cobalt from the mixed solution, resulting in the efficient extraction of cobalt metal and effective separation of cobalt from lithium. The research encompasses various aspects, including the pretreatment of waste LCO, the determination of optimal leaching conditions, characterization of leaching kinetics, exploration of the leaching process mechanisms, and subsequent electrochemical analysis of the leaching solution along with cobalt extraction via electrodeposition. This study aims to enhance the understanding of the practicality and environmental sustainability of recycling waste LCO, while also contributing to the rapidly evolving fields of battery recycling technology and waste management.

EXPERIMENTS

Materials

A battery recycling company provided waste lithium cobalt batteries. Sinopharm supplied citric acid ($C_6H_8O_7$-H_2O, 99%) and sodium thiosulfate ($Na_2S_2O_3$, 99%) for the citric acid leaching experiments, while Shanghai McLean Biochemistry and Technology Co. supplied ammonia (NH_3-H_2O, 99%), sodium hydroxide (NaOH, 99%), and sodium sulfate (Na_2SO_4, 99%) for the deposition experiments. The electrodeposition experiment used a 304 stainless steel cathode (20 × 13 × 1 mm) and a platinum sheet anode (20 × 15 × 1 mm), with a 70 mm distance between the two plates.

Experimental Procedures

The experimental approach is depicted in Fig. 1. To begin, pure $LiCoO_2$ powder was pretreated before proceeding to the next stage. The leaching mixed solution was then prepared by acidifying $LiCoO_2$ with citric acid and sodium thiosulfate. The cobalt metal in the solution was then removed by electrodeposition, resulting in lithium cobalt separation.

Pre-Processing

The recovered lithium cobaltate batteries were discharged in the specified NaCl solution for 48 h. After discharge, the battery was disassembled to obtain the entire positive electrode material, negative electrode material, and additional components. The dismantled cathode material was soaked in water and shaken with an ultrasonic device for 5–8 min. It was then dried, placed in a crucible, and roasted in a muffle furnace. The heat-treated lithium cobalt oxide powder was crushed and sieved and then kept in a dry condition for use as a leaching material.

Leaching Processes

To facilitate the leaching reaction, the pretreated material was dissolved in a citric acid solution and continuously agitated at 600 rpm. The study examined how citric acid concentration (0.4–1.4 mol/L), sodium thiosulfate concentration (0.15–0.4 mol/L), temperature (40–90 °C), and leaching period (30–80 min) affect the rate of leaching. The leaching rate was computed as Eq. 1.[60,61]

$$R = \frac{n \times C \times V}{m \times \omega} \times 100\% \quad (1)$$

In the equation, R was the metal element's leaching efficiency, n was the solution's dilution (n = 1000), C was the result of the ICP-OES test concentration, V was the volume of the leaching

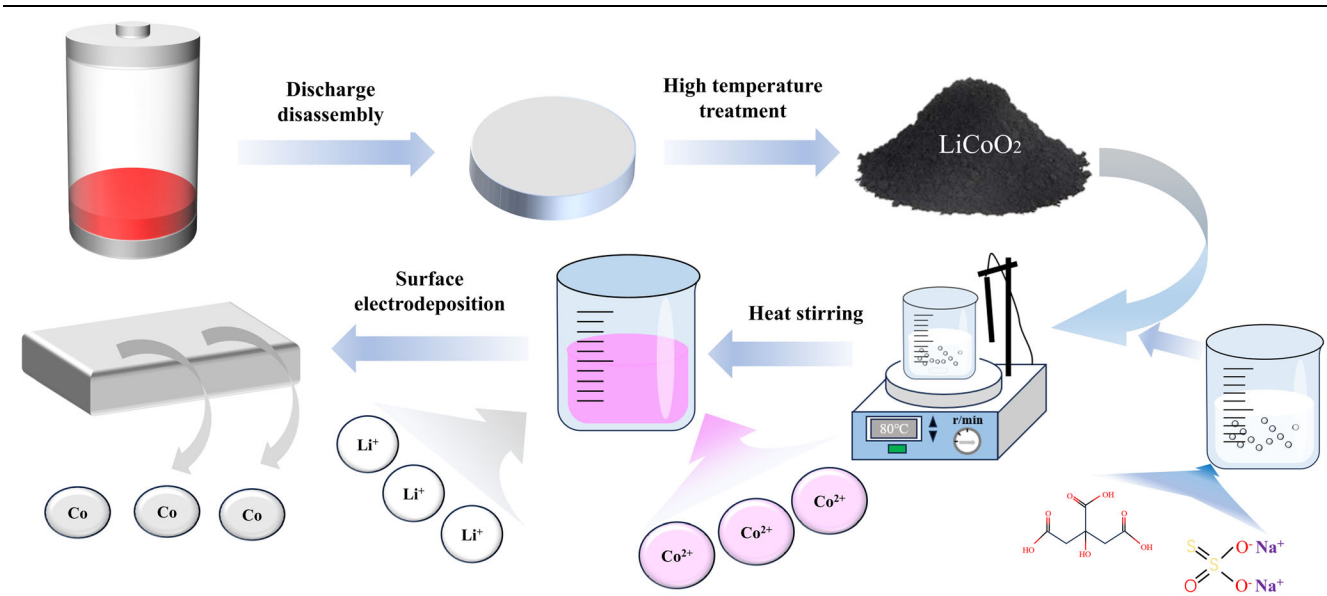

Fig. 1. Experimental procedures.

solution, m was the mass of the leaching agent, and ω was the mass fraction of metal in the material.

Electrodeposition Experiment

In the electrodeposition studies, the cathode steel plate made of 304 stainless steel was sanded and polished with 400–2000 grit sandpaper, then rinsed with deionized water. This procedure was designed to produce a smoother, flatter surface on the substrate, making it easier to apply subsequent plating. The electrodeposition experiments were performed after the treatment at a cobalt mass concentration of 42 g/L, an electrodeposition temperature of 63 °C, a pH of 4 in the electrodeposition solution, a current density of 450 A/m^2, a sodium sulfate mass concentration of 145 g/L, and an electrodeposition time of 180 min. The electrodeposition recovery of cobalt was determined using Eq. 2.

$$\theta = \frac{C_0V_0 - C_tV_t}{C_0V_0} \tag{2}$$

In the above equation, C_0 was the initial concentration of metal ions (g/L) in the leach solution prepared before electrodeposition; V_0 was the volume of leach solution used before electrodeposition (L); C_t was the concentration of metal ions remaining in the filtrate; V_t was the volume of filtrate.

Characterization and Analysis Methods

The link between enthalpy change and mass change at elevated temperatures during the high-temperature reaction of $LiCoO_2$ powder was investigated using a STA 449 F3 thermogravimetric analyzer (TG-DSC). The metal ions Co^{2+} and Li^+ were examined using the SPECTRO GENESIS inductively coupled plasma spectrometer (ICP-AES). A Pan Analytical X Pert Powder X-ray diffractometer (XRD) was used to analyze the physical phases of the heat-treated powders. The X-ray tube produced Cu Kα1 radiation with a wavelength (λ) of 0.15405 nm. The tube voltage was 40 kV, and the current was 20 mA. The scanning rate was set to 40º/min, and the range (2θ) was from 10° to 90°. The micro-morphologies of $LiCoO_2$ powder following high-temperature heat treatment and cobalt metal coatings generated by electrodeposition were examined using a Zeiss-IGMA HD scanning electron microscope. The leaching solution was tested using an electrochemical workstation. The linear scanning voltammetry test was performed over a pH range of 2.0–7.0, a plating solution temperature of 20–25 °C, a scanning range of − 2.0–2.0 V/SCE, a scanning rate of 50 mV per second, and a starting potential of − 2.0 V/SCE. The cyclic voltammetry test was performed in the pH range of 3.0–7.0, with a scanning range of − 3.0–3.0 V/SCE, a scanning rate of 50 mV per second, and a starting potential of 0 V/SCE.

RESULTS AND DISCUSSION

Thermal Treatment Results

The cathode material sheet was dried and analyzed using a thermogravimetric-differential thermal synthesizer (TG-DSC). As illustrated in Fig. 2a, the DSC curve exhibits two significant exothermic peaks. The two weight loss intervals indicated by the TG curve correspond to the temperatures of the exothermic peaks observed in the DSC curve. The exothermic peak at 460 °C indicatess the degradation of the binder polyvinylidene fluoride (PVDF), while the peak at 600 °C corresponds to the oxidation process of acetylene black.[26,62] The initial sample mass for the experiment was 9.7 mg, with a mass loss of 0.404 mg, resulting in a final residual mass percentage of 95.83%. The first high-temperature heat treatment was conducted at 500 °C for 3 h, during which the PVDF on the collector foil was effectively removed, allowing for the physical peeling of the residual material due to the ease of removing the active material. The second high-temperature treatment, lasting 4 h at 600 °C, nearly eliminates PVDF and acetylene black, significantly reducing the fraction of impurities in the cathode material and greatly enhancing the quality of the cathode active material. Inductively coupled plasma atomic emission spectroscopy (ICP-AES) was employed to determine the concentrations of metal ions Co^{2+} and Li^+, revealing that the synthesized lithium cobaltate powder contained no metal impurities other than cobalt and lithium.

As shown in Fig. 2b, the comparative analysis indicates that the principal diffraction peaks remained constant, with no additional peaks identified. Furthermore, the positions of these diffraction peaks were in good agreement with the crystal plane positions specified on the $LiCoO_2$ standard card. This behavior suggests that the $LiCoO_2$ powder recovered and synthesized from waste lithium-ion batteries is of high purity, and the heat treatment method did not introduce impurities or alter its basic crystal structure.

As illustrated in Fig. 3, the $LiCoO_2$ powder, after the second high-temperature heat treatment, exhibits a homogeneous particle distribution and regular particle morphology. The particle size distribution closely resembles that of pure $LiCoO_2$ powder, as does the particle shape. This similarity indicates that the recovered and heat-treated $LiCoO_2$ powder can be utilized directly in subsequent procedures without the need for additional purification stages.

Effect of Reaction Parameters on Leaching Efficiency

Figure 4 illustrates the impact of four parameters in the citric acid-sodium thiosulfate system on the leaching rates of Co^{2+} and Li^+: (1) concentration of citric acid, (2) concentration of $Na_2S_2O_3$, (3)

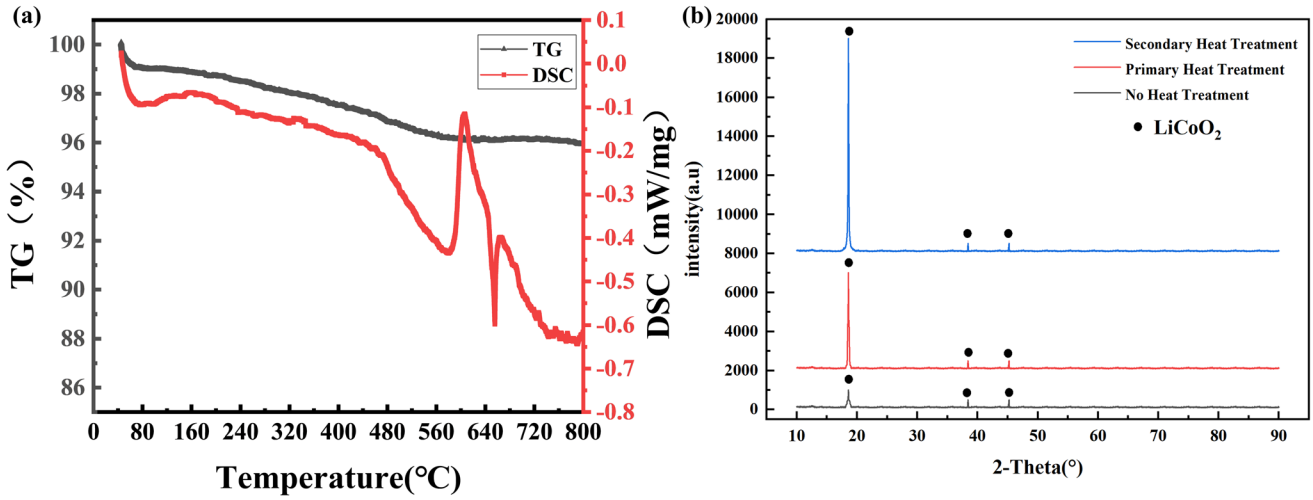

Fig. 2. (a) TG-DSC test pattern of battery cathode material. (b). XRD diffraction pattern of battery cathode material.

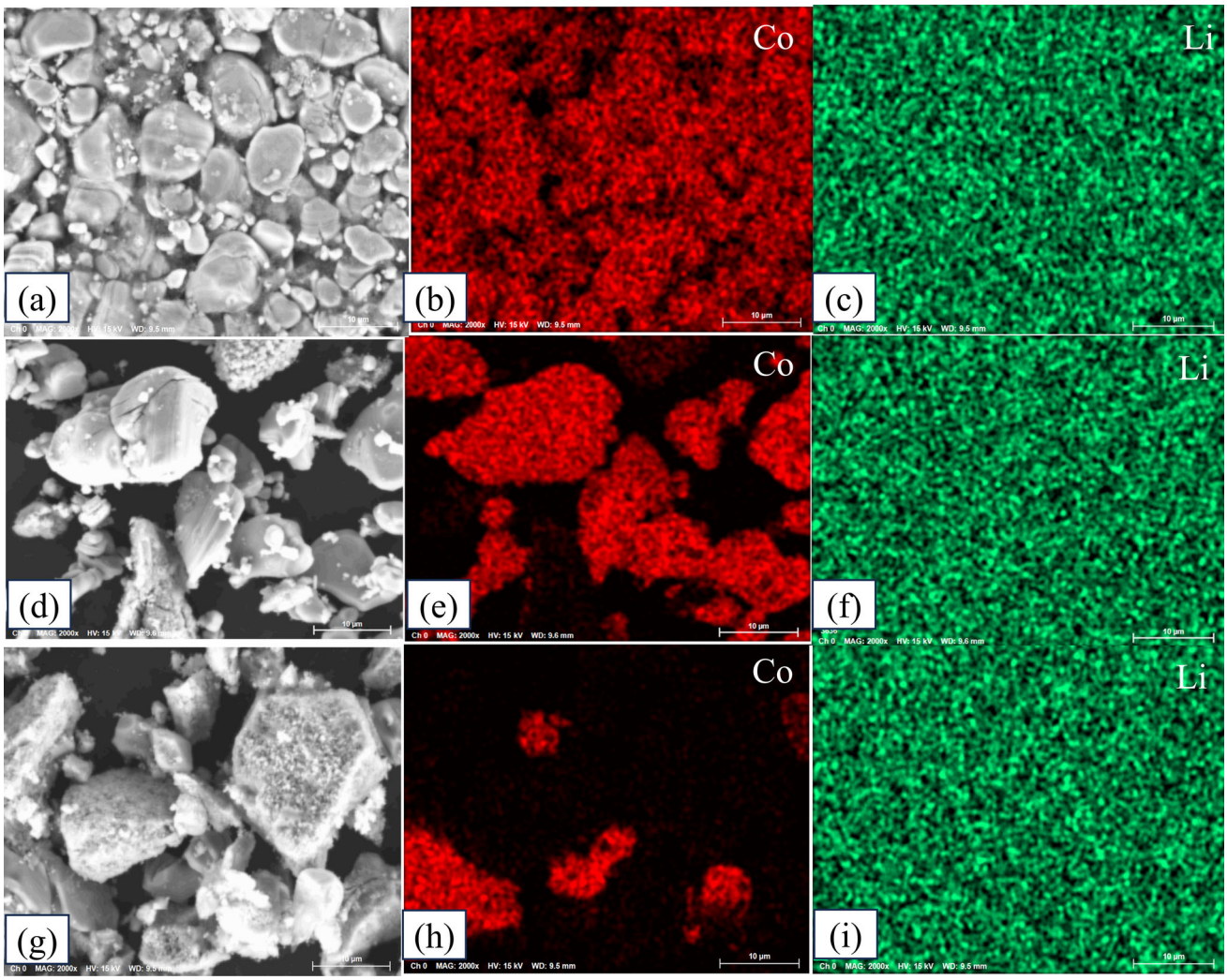

Fig. 3. Morphological changes in waste cathode materials prior to and following high-temperature treatment: (a, b, c) waste cathode materials. (d, e, f) cathode materials after primary high-temperature treatment; (g, h, i) cathode materials after secondary high-temperature heat treatment.

temperature, and (4) leaching time. The optimal parameters for the citric acid-sodium thiosulfate system were identified as follows: a citric acid concentration of 1.2 mol/L, a $Na_2S_2O_3$ concentration of 0.35 mol/L, a temperature of 80 °C, and a leaching time of 70 min.

(a) Citric acid concentration

The leaching rates of Co and Li metals showed a direct correlation with the citric acid concentration. A maximum leaching rate of 97.5% was obtained at a citric acid concentration of 1.2 mol/L. Beyond this concentration, the H^+ ion concentration in the solution decreased, while the disparity in H^+ concentration between the solid-liquid interface and the solution increased, thereby accelerating the diffusion rate of the leaching reaction and facilitating the leaching process. Consequently, a reduction in the leaching rates of Co and Li metal ions was observed. Moreover, as the citric acid concentration continued to rise, the leaching efficiency of Co and Li metal ions did not increase; instead, it declined. This phenomenon can be attributed to the fact that, with the increasing citric acid concentration, the acid leaching reaction approached an equilibrium state, resulting in a constant diffusion rate and, consequently, no significant increase in the amount of leached metal ions. When the citric acid concentration reached 1.4 mol/L, the leaching rates of Co and Li metals decreased during the leaching process. This decline is due to the increased concentration of H^+ ions, which, at excessively high levels, significantly inhibits the formation of H_2Cit^-, $HCit^{2-}$, and Cit^{3-} complexes.[63] This inhibition hampers the detachment of metal ions from the solid phase, leading to the formation of stable complexes that hinder their entry into the solution, ultimately resulting in a reduced leaching rate.

(b) The concentration of $Na_2S_2O_3$

The efficiency of Co and Li metal ion leaching increases with rising concentrations of $Na_2S_2O_3$. In the citric acid-sodium thiosulfate leaching system, $Na_2S_2O_3$ primarily functions as a reducing agent, ensuring the complete reduction of Co^{3+} to Co^{2+} during the leaching reaction. The main objective of adding sodium thiosulfate is to facilitate the conversion of Co^{3+} to Co^{2+}. However, when the $Na_2S_2O_3$ concentration exceeds 0.3 mol/L, undesirable side reactions occur within the leaching system. This phenomenon is attributed to the excessive concentration of $Na_2S_2O_3$, which leads to the formation of a hydrophobic film

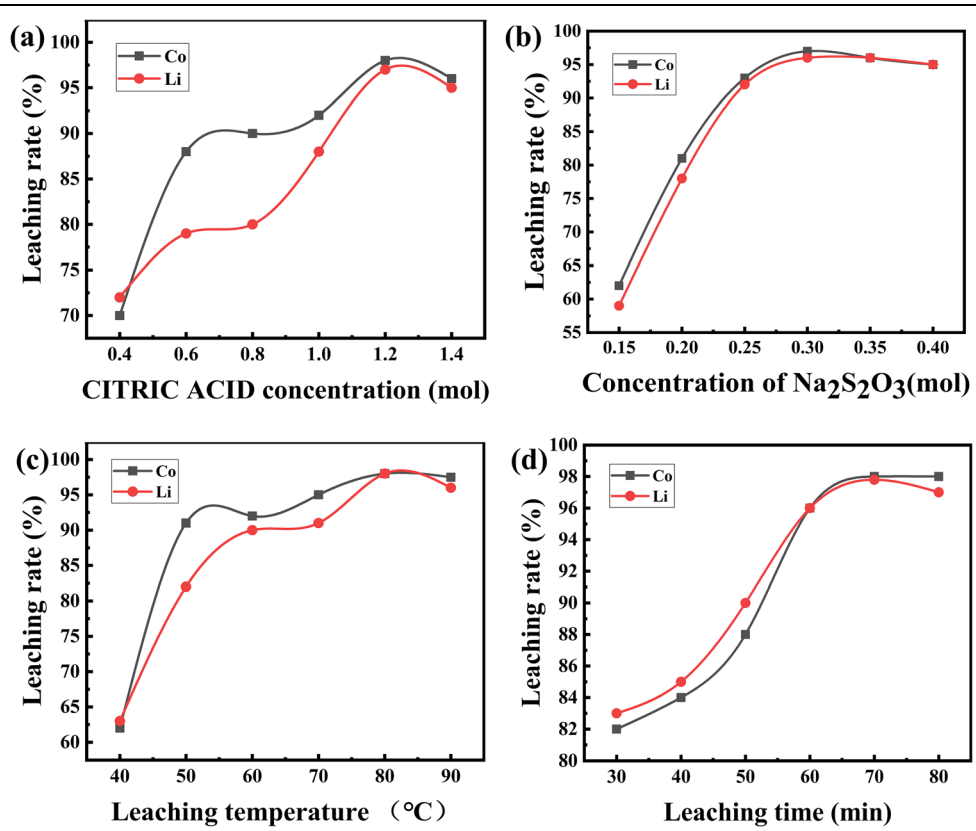

Fig. 4. (a) Citric acid concentration; (b) sodium thiosulfate concentration; (c) temperature; (d) holding time. Conditions: heating rate 5 °C/min.

from sulfur monomers generated during the reaction. This film physically obstructs the surface of the material, diminishing the contact area between the $LiCoO_2$ powder and the solvent, thereby impeding the leaching reaction and causing a decline in the leaching rates of Co and Li metal ions.

(c) Temperature

At 40 °C, the leaching rates of cobalt (Co) and lithium (Li) metals were approximately 62% and 63%, respectively. The leaching efficiency of Co and Li metal ions increased significantly with rising temperature. The highest leaching rates for Co and Li were observed at 80 °C, reaching approximately 98%. As illustrated in Fig. 4d, extending the leaching duration to 70 min maximized the leaching efficiency.

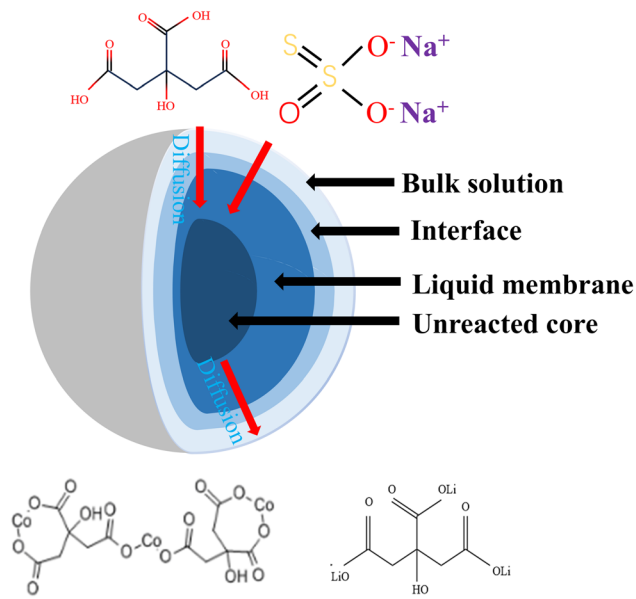

Fig. 5. Schematic of the leaching kinetic mechanism.

Dynamics Analysis

Metal ion leaching curves and kinetics were investigated at temperatures ranging from 60 °C to 80 °C to enhance the understanding of the underlying reaction mechanisms.

Metal leaching in acidic solutions is generally characterized as a non-homogeneous solid-liquid interaction that primarily occurs at the surface of metal particles.[64] During the leaching process, metal ions are continuously released from the surface of the solid particles, resulting in a gradual contraction of the reaction interface toward the particle core. The remaining solid material subsequently forms a gray, structurally loose, porous layer, allowing the leaching products to permeate back into the original liquid.[65] This phenomenon aligns with the principles of the shrinking core model (Fig. 5).

Typical approaches for this model include surface chemical control (3), internal diffusion control (4), and hybrid control models (5), as illustrated in the mathematical Eqs. 3, 4, 5, and 6.[51,66]

$$1-(1-x)^{\frac{1}{3}}=k_1t \tag{3}$$

$$1-\frac{2}{3}x-(1-x)^{\frac{2}{3}}=k_2t \tag{4}$$

$$\ln\left(1/3\ln(1-x)-1+(1-x)^{-1/3}\right)=\ln k_3+\ln t \tag{5}$$

$$\ln k=-\frac{\mathrm{Ea}}{\mathrm{RT}}+\ln A \tag{6}$$

In the equation above, k was the leaching reaction rate constant, x was the leaching efficiency, t was the leaching duration, A was the pre-exponential factor, Ea was the apparent activation energy, R was the gas constant, and T was the reaction temperature.

Fitting the leaching kinetics data with the three equations presented above revealed that the hybrid control model for reaction Eq. 5 provided the best fit for the leaching process.

As illustrated in Fig. 6, the determination coefficients (R^2 values) for this model were > 0.96, indicating that the leaching process was primarily governed by the interplay between surface chemical reactions and diffusive mass transfer processes. The apparent activation energy was subsequently calculated using Arcadius equation 6. The correlation coefficients obtained from the fitting for cobalt and lithium were 0.907 and 0.975, respectively, with apparent activation energies of 31.53 kJ/mol and 46.98 kJ/mol. The observed decrease in the fit of the limiting link with increasing temperature may be attributed to the reduced solubility of the citric acid-sodium thiosulfate system in aqueous solution, which consequently leads to a decrease in the molecular weights of citric acid and sodium thiosulfate in the leachate. This reduction in molecular weight results in a diminished reaction rate of the native solution at elevated temperatures, allowing the limiting link to occur during the mass transfer step.

The apparent activation energy of lithium (Li) was relatively high, making ion leaching more challenging. This phenomenon is primarily due to the weak bond between Li and citrate, which decomposes at elevated temperatures. Consequently, along with a reduction in the Li metal content of the waste lithium battery cathode material, the initial step of the surface chemical reactions is hindered. In contrast, cobalt (Co) leaching

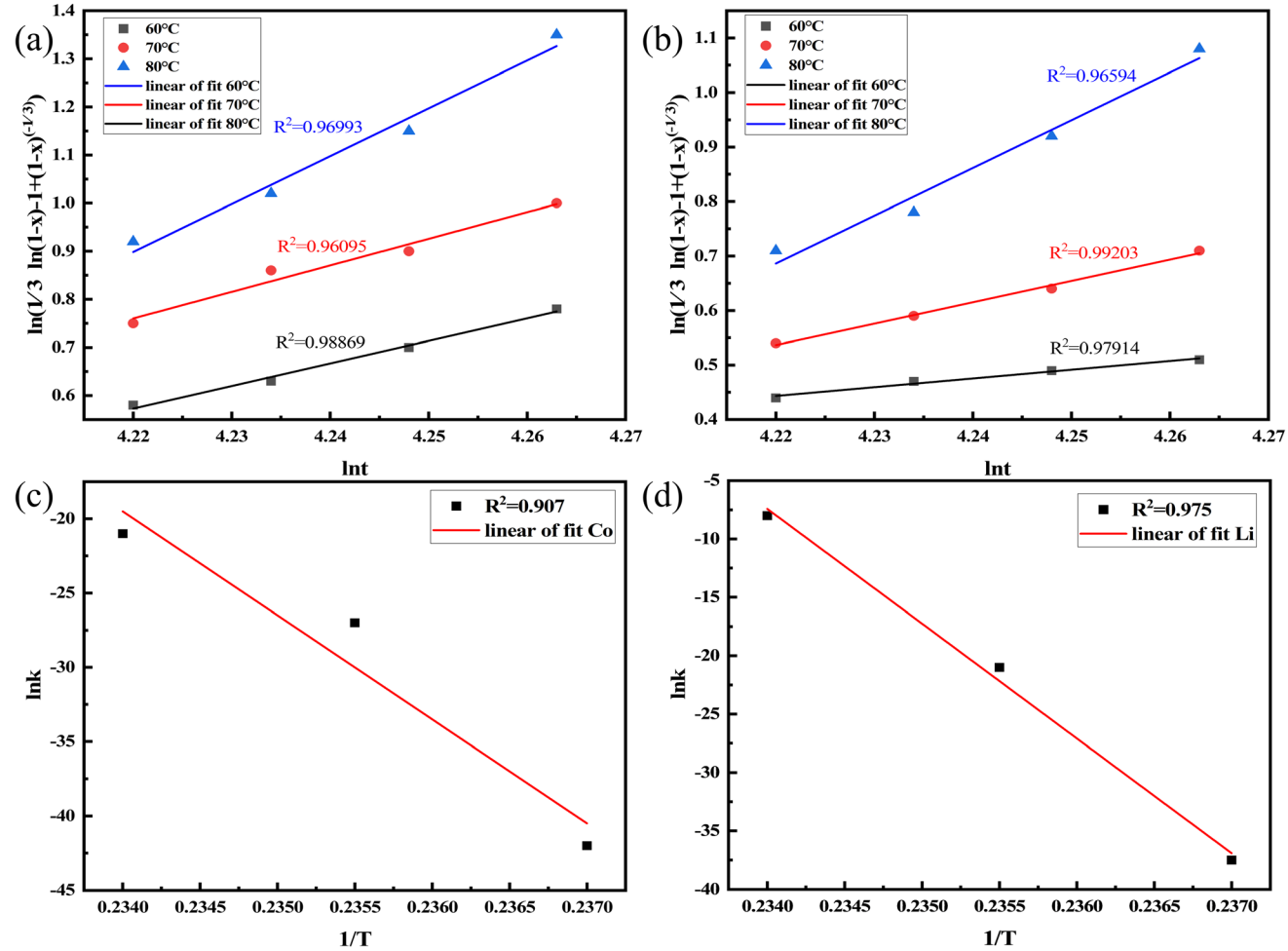

Fig. 6. The linear correlation associating the expression ($\ln(1/3\ln(1\text{-}x)\text{-}1+(1\text{-}x)^{-1/3})$) with the lnt was observed for the elements (a) Co, (b) Li; Arrhenius equation fitting curves for the hybrid control model of (c)Co, (d) Li.

was comparatively significant, obstructing the surface chemical processes of Li and limiting the interactions within the mass transfer process. This obstruction necessitates a larger activation energy for Li leaching.[67,68]

DFT Analysis

As shown in Fig. 7a, the stable range for Co^{2+} ions in acidic solutions is $-0.3\ V < E < 1.25\ V$. As the potential increases, Co^{2+} ions become stable in a more acidic environment. It is believed that when $E < -0.3\ V$ and $pH \leq 3$, Co^{2+} predominantly exists in the free state in the citric acid leach solution. Conversely, when the potential decreases and $E > 0.5\ V$, a minor amount of $Co(C_6H_6O_7)$ is formed in the citric acid solution. It has been hypothesized that when $-1.5\ V < E < -0.5\ V$ and $4 \leq pH \leq 6$, Co^{2+} in the citric acid leach solution primarily exists as $Co(C_6H_5O_7)^-$ and is readily reduced. When the potential E exceeds 1.25 V, Co^{2+} ions in the solution cannot remain stable and will be oxidized to Co^{3+} ions. Under low-potential conditions, $Co(OH)_2$ can remain stable in alkaline solutions with $pH > 9.0$. Li^+ ions can remain stable in solution at $pH < 7.0$, but at $pH > 9.0$, they form hydroxides.

According to the E-pH diagram, the chemical reaction equations involved in the leaching of cathode materials from waste lithium cobalt oxide (LCO) are shown in the citric acid-sodium thiosulfate system. During the leaching reaction, citric acid undergoes tertiary ionization, leading to the leaching of lithium cobaltate, as illustrated in Fig. 7b. Citric acid is a powerful acid capable of ionizing three protons ($3H^+$) in solution. To ensure charge conservation and prevent the carboxyl group (COO^-) from interacting with nonmetal ions during computations, citric acid was modeled in its molecular form by incorporating hydrogen atoms at the carboxyl terminal. Upon interaction with cobalt ions, the capped hydrogen was released. Citric acid exhibits coordination and complexation properties; therefore, before coordinating with cobalt, it is essential to analyze the orbital information of citric acid to identify the binding site for cobalt.[69] Figure 7c, d, e, and f displays the optimized structure of citric acid (CA) along with its highest occupied

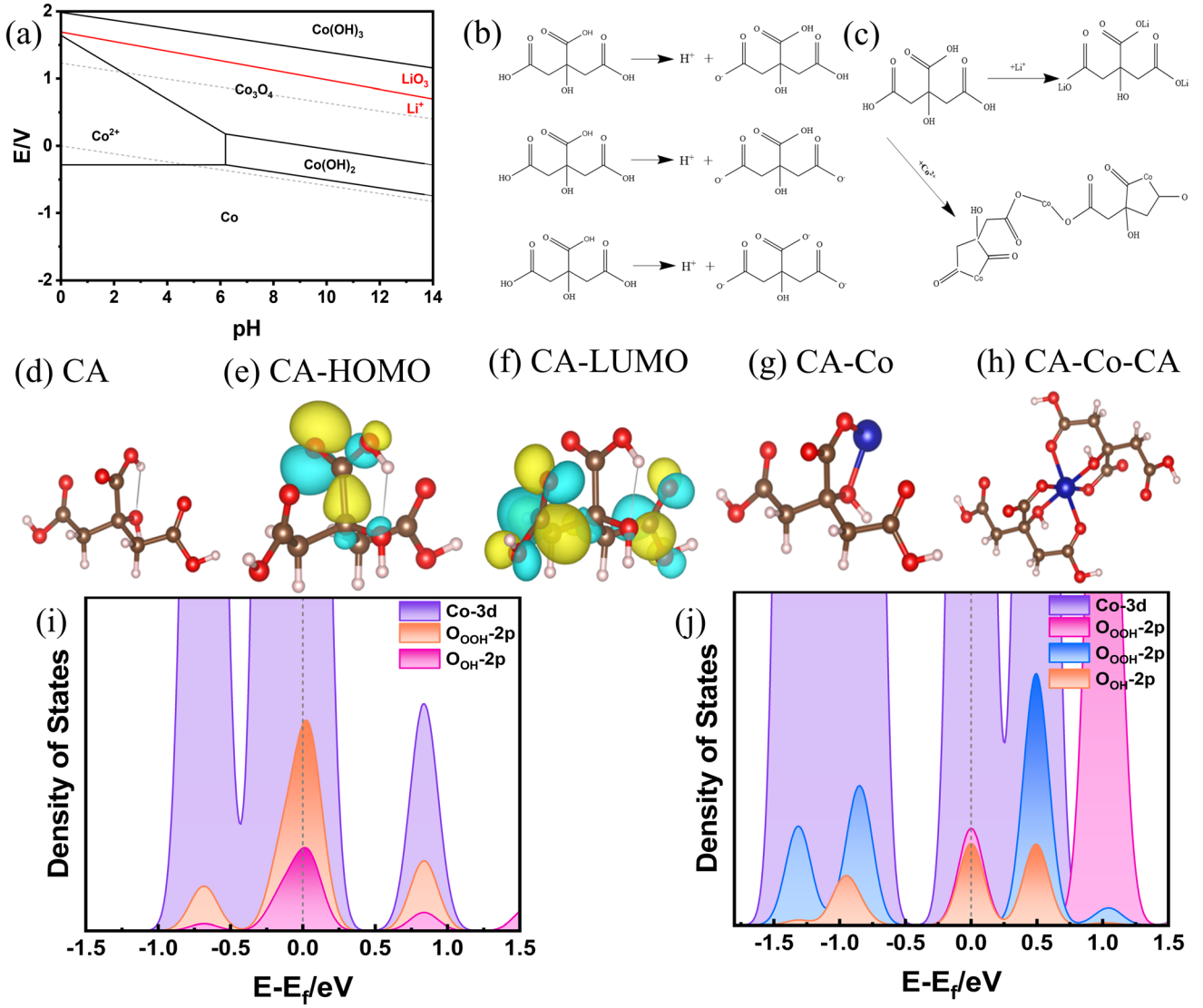

Fig. 7. (a) Potentiometric pH diagram of Co-H_2O, Li-H_2O system (temperature 298.15 K, pressure 1.01 kPa). (b) Ionization equation for citric acid in aqueous solution. (c) Reaction equation of citric acid-sodium thiosulfate system with lithium cobaltate. (d) CA (citric acid). (e) CA-HOMO (highest occupied orbital). (f) CA-LUMO (lowest occupied orbital). (g) CA-Co. (h) CA-Co-CA. (i) Density of states of CA-Co. (j) Density of states of CA-Co-CA.

molecular orbital (HOMO) and lowest unoccupied molecular orbital (LUMO).

The electrons in the highest occupied orbital of CA were primarily localized on the carboxyl group, indicating that this group has a lower binding capacity for electrons and functions as an electron donor, making it particularly sensitive to electron loss. In contrast, the electrons in the lowest occupied orbital were predominantly centered on CA's terminal carboxyl group, which exhibits a superior electron binding capacity and acts as an electron acceptor, facilitating the easy acquisition of electrons. Based on the characteristics of CA, it can be concluded that cobalt (Co), in its ionic form, is already in a stable state in solution, rendering it highly improbable for Co to lose electrons and transition to a higher valence state. When coordinating with CA, Co is expected to preferentially interact with the carboxyl group of the electron donor, transferring electrons to the Co ion through the carboxyl group. This interaction allows the oxygen atom on the carboxyl group to share a portion of its electrons with Co, resulting in the formation of a stable chemical bond. Consequently, there are at least two types of coordination between CA and Co: Co bonded to CA (CA-Co) (Fig. 7g) and Co bonded to two CAs (CA-Co-CA) (Fig. 7h).

The electrons in the highest occupied orbital of citric acid (CA) were primarily concentrated on the carboxyl group, indicating that in the complex CA-Co, cobalt (Co) was coordinated to both the carboxyl and hydroxyl groups on the branched chain. Given that the calculations above show no electron-losing capacity at the hydroxyl group, it is possible that

CA-Co may not exhibit structural stability. The structure of CA-Co-CA, in which Co is simultaneously coupled to multiple hydroxyl and carboxyl groups, could be even less stable and therefore warrants further investigation. In the solution system, the stability of the complexes is likely lower than that of the free ions; thus, relying solely on structural stability for determination may lead to incorrect conclusions. It is more significant to utilize chemical stability, as indicated by the HOMO-LUMO gap, for this assessment.[64,70] The computed gaps were 0.1279 eV for CA-Co and 0.492 eV for CA-Co-CA, suggesting that CA-Co-CA is more chemically stable. Meanwhile, CA's gap was 4.5797 eV, significantly higher than those of CA-Co and CA-Co-CA, indicating that the citrate ion is more stable. This also suggests that the bonding of Co with the hydroxyl group is very weak, resulting in only the bonding of Co with the carboxyl group in CA-Co-CA.

To further investigate the bonding capabilities of cobalt (Co) and carboxyl groups, density of states (DOS) calculations were performed, with the findings illustrated in Fig. 7i and j. The graphic clearly indicates that the bonding between Co and oxygen (O) is predominantly governed by electron hybridization in the Co-3d and O-2p orbitals, particularly near the Fermi energy level ($E - Ef = 0$ eV). The density states of Co and O bridge the Fermi energy level, demonstrating that the complexes exhibit conductive properties. Furthermore, Co shows a significant overlap and hybridization with hydroxylated O; however, hydroxylated O possesses a lower bonding capability compared to carboxylated O. The overlap and hybridization between Co and hydroxylated O are also substantial. Consequently, in solution, Co predominantly exists in the form of CA-Co-CA, and larger molecular clusters can be formed using this structural unit,[69] which aligns with the established coordinating ability of citric acid.

Extraction of Cobalt Metal

Electrochemical Analysis

Due to the solutions's high acidity, direct extraction via electrodeposition was not feasible. Therefore, it was necessary to modify the composition of the leaching solution. This study investigates the electrochemical behavior of the electrode surface through a series of experiments to identify optimal pH conditions that enhance electrodeposition quality.

As illustrated in Fig. 8, when pH ≤ 3.0, the linear-scan voltammetry curve of the leaching solution showed a reduction peak during cathodic scanning, with the peak potential approximately − 0.6 V/SCE. At this pH level, cobalt in the leaching solution predominantly exists as Co^{2+}, and the observed reduction peak corresponds to the reduction of Co^{2+} to cobalt atoms. In the pH range of 4–5, cathodic scanning revealed a reduction peak at − 1.2 V/SCE, whereas the reduction peak at − 0.6 V/SCE was no longer observed. At this stage, cobalt mainly existed as $Co(C_6H_5O_7)^-$; however, a small amount of Co^{2+} was still present. The citric acid solution appeared to inhibit the reduction of Co^{2+}, suggesting that the combined effects of citric acid and pH may have diminished the reduction process or rendered it less significant. Consequently, the reduction peak observed during this phase was primarily attributed to the reduction reaction of $Co(C_6H_5O_7)^-$ in solution.

It was speculated that the reduction process consists of two steps: first, $Co(C_6H_5O_7)^-$ is reduced to $Co(C_6H_6O_7)$; subsequently, $Co(C_6H_6O_7)$ is further reduced to Co. When pH ≥ 6, the scanning potential decreased from − 1.2 V to − 1.3 V as pH increased, leading to stronger cathodic polarization. When pH > 7, many dense bubbles appeared in the cathode area, signaling the start of the hydrogen precipitation reaction. As the reaction progressed, hydrogen precipitation became the primary cathodic reaction, the solution environment became alkaline, the citric acid and cobalt complex became more stable, and it was difficult to reduce Co^{2+} from solution.

It was speculated that the reduction process consists of two steps: first, $Co(C_6H_5O_7)$ is reduced to $Co(C_6H_6O_7)^-$; subsequently, $Co(C_6H_6O_7)^-$ is reduced to Co. When the pH is ≥ 6, the scanning potential decreases from − 1.2 V to − 1.3 V as the pH increases, indicating stronger cathodic polarization. At pH values > 7, a significant number of dense bubbles appears in the cathode area, indicating the onset of the hydrogen precipitation reaction. As the reaction progresses, hydrogen precipitation becomes the primary cathodic reaction. The solution environment becomes alkaline, causing the citric acid and cobalt complex to stabilize, making it difficult to reduce Co^{2+} from the solution.

As shown in Fig. 8, two peaks were observed in the cyclic voltammetry scanning curve at pH = 3, with onset potentials of − 0.6 V/SCE and − 1.4 V/SCE. The hydrogen precipitation reaction occurred in the cathodic region at − 1.4 V/SCE. It is hypothesized that − 0.6 V/SCE corresponds to the reduction peak of Co^{2+}, which is reduced to cobalt atoms at this potential, primarily in the form of free Co^{2+} ions. When the pH ranged from 4 to 5, the cyclic voltammetry cathodic scanning curve exhibited two reduction peaks at − 0.6 V/SCE and − 1.36 V/SCE. At − 0.6 V/SCE, it is assumed that a small amount of Co^{2+} in the solution underwent a reduction reaction, while at − 1.36 V/SCE, this peak corresponds to the reduction of $Co(C_6H_5O_7)^-$ in the solution. At this point, the reduction potential was significantly more negative than the linear voltammetry test potential, indicating a more pronounced cathodic polarization, during which Co^{2+} was reduced. At pH > 6, the cyclic voltammetric curves did not show any reduction peaks, indicating hydrogen precipitation in solution, which is largely consistent with the results from linear scanning

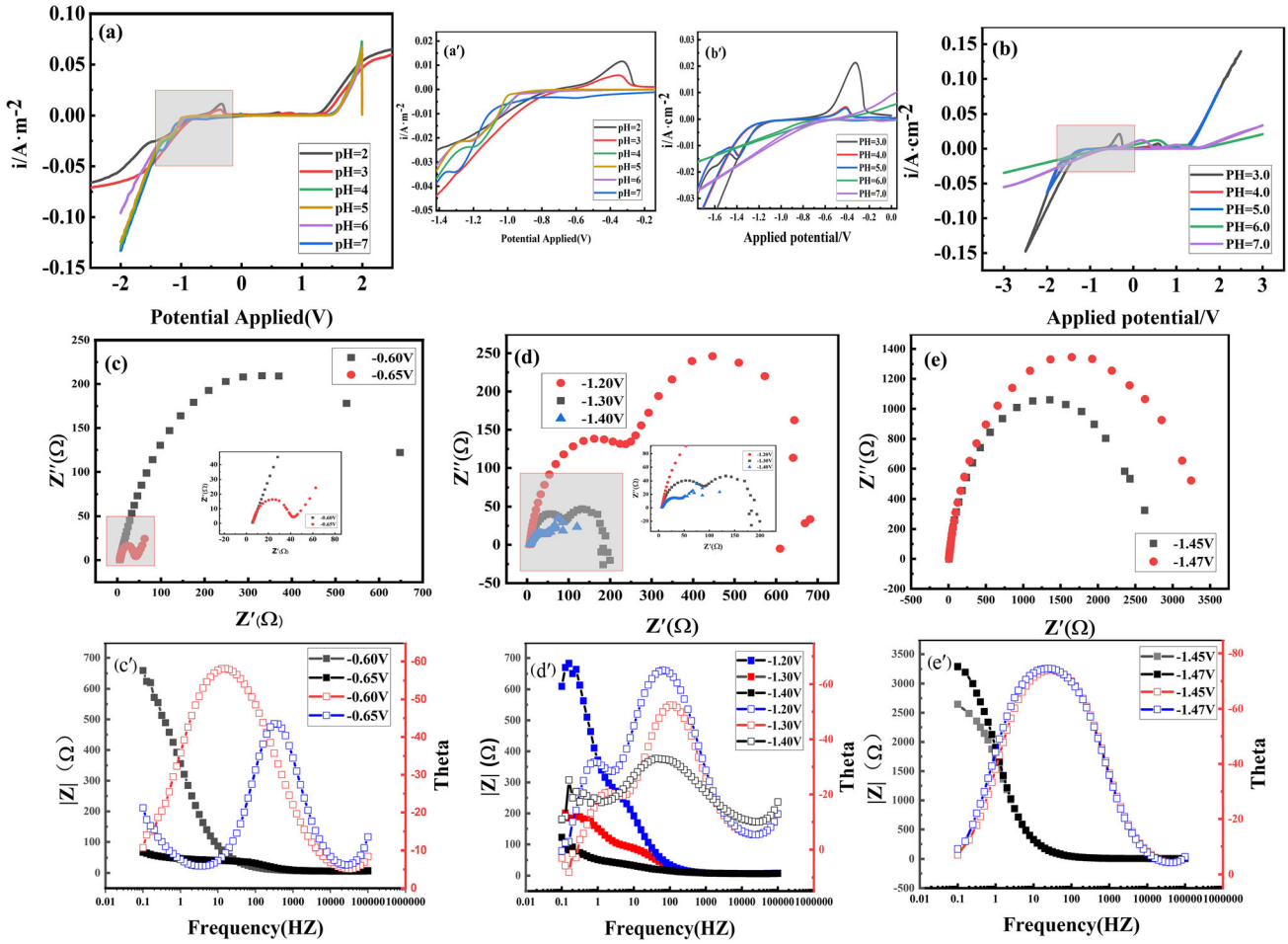

Fig. 8. (a). Linear scanning voltammetry curves for different pH deposition solutions (a') as a local magnification (b). Cyclic voltammetry curves for different pH deposition solutions (b') were a localized magnification ((c) (d) (e) (c') (d') (e')). AC impedance spectra of Co^{2+} at different potentials at pH = 4.

voltammetry tests. When 4 ≤ pH ≤ 5, the reduction potential of Co^{2+} on the cyclic voltammetry curve exhibits a significant negative shift.

The AC impedance test was conducted using the potential range of the reduction peaks observed in the linear and cyclic voltammetric test curves at pH = 4. As illustrated in Fig. 8, at an applied potential of − 0.60 V, the Nyquist plot exhibits a prominent capacitive reactance arc, while the corresponding Bode plot indicates a reduction peak. This observation suggests that a single electrochemical reduction reaction occurs at − 0.60 V/SCE, characterized by relatively high resistance. Upon adjusting the applied potential negatively from − 0.60 V/SCE to − 0.65 V/SCE during the AC impedance test, the Nyquist plot reveals a capacitive resistance semicircle with a diameter of approximately 40 in the high-frequency region, accompanied by a straight line in the low-frequency zone. At this juncture, the Bode plot also displays a reduction peak, indicative of Co^{2+} reduction, which is a one-step reduction process potentially occurring with reduced resistance.

Based on the AC impedance spectra, it is hypothesized that the Co^{2+} reduction process at this stage is influenced by a combination of kinetic and diffusion controls. The straight-line peaks observed in the low-frequency region are attributed to the transition from finite to infinite diffusion of the chemical reactants.[71,72] At an applied potential of − 1.20 V/SCE, the Nyquist plot reveals a radial capacitive reactance curve in the high-frequency region and a semicircular inductive reactance arc in the low-frequency region. At a potential of − 1.30 V/SCE, the Nyquist plot presents a semicircle of capacitive reactance in the high-frequency zone and a semicircle of susceptibility in the mid-frequency range. The Nyquist figure illustrates a section of capacitive reactance arcs in the high-frequency area at a lower potential (− 1.40 V/SCE) and a section of susceptibility arcs in the middle frequency region. The presence of susceptibility arcs in the Nyquist plots indicates that Co^{2+} in solution was reduced, resulting in an intermediate product that appears to be adsorbed on the electrode surface.[73]

During the negative shift of the reduction potential from − 1.20 V/SCE to − 1.40 V/SCE, two reduction peaks appeared in the Bode diagram. This observation suggests that the Co^{2+} reduction process occurs via a two-step reaction: the first intermediate product is adsorbed onto the surface of the electrode, followed by its reduction to the atomic state. At this stage, the intermediate product is reduced to its atomic form, marking a crucial phase in the regulation of the Co^{2+} reduction process. As the negative potential is applied, the Co^{2+} reduction reaction becomes more challenging, accompanied by a mild hydrogen precipitation reaction. At applied potentials of − 1.45 V and − 1.47 V/SCE, the Nyquist plots reveal a broad capacitive resistance semicircle with only one reduction peak observed in the Bode diagram. Further analysis indicates that this reduction peak corresponds to the Co^{2+} reduction reaction in the solution, which is a one-step process. When the potential is negatively shifted from − 1.45 V/SCE to − 1.47 V/SCE, the diameter of the capacitive resistance semicircle in the Nyquist diagram decreases, indicating a reduction in the resistance of the reduction reaction in the solution. It is posited that the Co^{2+} reduction reaction in the solution is challenging at this point, with the hydrogen precipitation reaction being the dominant process.

When the applied potential is more negative than − 1.45 V/SCE, the hydrogen precipitation reaction in the solution becomes intense, complicating the reduction of Co^{2+} and making it difficult for the reduction reaction to reach completion. The findings from this section of the analysis are generally consistent with the results obtained from the cyclic voltammetry tests. It was determined that the optimal pH for the electrodeposition reaction is 4.

Surface Morphology of Cobalt Coatings

The optimal conditions for cobalt electrodeposition were determined through a series of experiments. These conditions include a cobalt mass concentration of 42 g/L, an electrodeposition temperature of 63 °C, a solution pH of 4, a current density of 450 A/m^2, a sodium sulfate mass concentration of 145 g/L, and an electrodeposition duration of 180 min.

Metal electrodeposition was used to extract cobalt metal ions from the leaching solution, which were subsequently deposited as a monomer on a cathodic stainless steel plate substrate. The microscopic and macroscopic morphology of the plated layer was analyzed using a scanning electron microscope, as illustrated in Fig. 9a.

As shown in Fig. 9b, the coating exhibits a metallic luster, with a very uniform distribution of cobalt monomers on the substrate. The outer surface is smooth, and the cobalt atoms deposited on the stainless steel substrate are tightly aligned, with no gaps. The distribution of surface atoms is uniform and flat, with no pits, pitting, or scratches. Similarly, Fig. 9c illustrates that the coating retains a metallic luster, with a consistent distribution of cobalt monomers on the substrate; the outer surface remains smooth, and the cobalt atoms deposited on the stainless steel substrate are tightly aligned, with ni gaps. The surface atom distribution is uniform and flat, free from pits, pitting, or scratches. Furthermore, the cobalt metal atoms within the stainless steel substrate are densely packed, with consistent atomic growth in both form and direction. The shape resembles a prismatic cone, with the cone tip oriented toward external growth, ensuring complete coverage of the stainless steel surface and resulting in excellent densification.

Cobalt Coating Electrodeposition Recovery Efficiency Analysis

Cobalt and lithium can be effectively leached from lithium cobaltate powder using a citric acid-sodium thiosulfate system with optimal parameter adjustments. The leach solution was used as a plating solution, and cobalt metal was recovered through electrodeposition studies. Meanwhile, the residual lithium metal was converted into lithium hydroxide via membrane electrolysis.

The cobalt recovery was assessed after three electrodeposition cycles, revealing that approximately 90% of cobalt was recovered in the form of

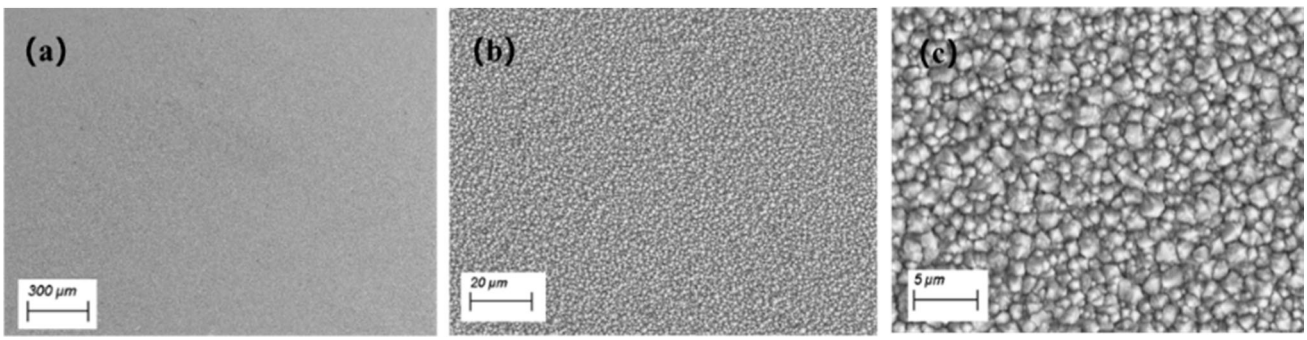

Fig. 9. Macroscopic morphology of cobalt coating: (a) 30X, (b) 500X, (c) 2000X.

Table I. Co^{2+} concentration in solution and weight gain of substrate after three depositions

	Original solution	Primary deposition	Secondary sedimentation	Tertiary sedimentation
Concentration of Li^{+} in solution after deposition (g/L)	0.271	0.264	0.258	0.253
Concentration of Co^{2+} in solution after deposition (g/L)	2.445	0.363	0.158	0.063
Weight gain of substrate after deposition (g)	0	2.77	0.132	0.003

a plating layer. The details of the cobalt recovery are presented in Table I below.

CONCLUSION

In this study, lithium cobalt oxide powder was initially separated from the raw material using a high-temperature heat treatment procedure. Subsequently, a reductive leaching technique using citric acid and sodium thiosulfate was developed to facilitate the effective leaching of cobalt and lithium metals from waste lithium cobalt oxide (LCO). This method enabled the deposition of cobalt metal by electrodeposition, thereby separating cobalt and lithium metals from a mixed solution of cobalt and lithium citrates. Under optimal leaching conditions, the leaching efficiencies for both cobalt and lithium exceeded 98%. Kinetic investigations indicated that the leaching of cobalt and lithium from the cathode materials of waste LCO was primarily constrained by the combined effects of surface chemical reaction kinetics and diffusive mass transfer mechanisms. In this research, the reaction mechanism of the leaching process was thoroughly examined, revealing that cobalt predominantly existed in the chemical forms of cobalt citrate (CA-Co) and cobalt citrate dimer (CA-Co-CA) during the reaction. Based on studies of chemical stability and density-of-states function calculations, it was concluded that cobalt primarily exists as cobalt citrate dimer in solution. Electrochemical studies were conducted to identify the optimal deposition conditions for the solution, facilitating the electrodeposition of cobalt metal onto the plated layer. Experimental results demonstrated that the extraction efficiency of cobalt metal can reach ≥ 90%, enabling effective separation of cobalt and lithium in solution.

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DATA AVAILABILITY

Data will be made available on request.

CONFLICT OF INTEREST

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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